

ABSTRACT: C++ PASSWORD GENERATOR & EVALUATOR

1. Overview

In an increasingly digitized landscape, robust password security remains the fundamental barrier against unauthorized access and data breaches. Many users rely on predictable, easily guessable passwords, compromising system integrity. This project presents a lightweight, console-based Password Generator and Strength Evaluator developed in C++. The dual-purpose utility empowers users to autonomously generate cryptographically resilient passwords while simultaneously evaluating the strength of the generated strings based on heuristic complexity rules.

2. Methodology and Implementation

The utility is engineered using standard C++ libraries, primarily leveraging string manipulation and time-seeded pseudo-random number generation. The core logic is bifurcated into two primary modules:

Password Generation: The generation engine utilizes user-defined constraints—specifically desired length and character set preferences (inclusion of numerals and special symbols). Based on these parameters, the program dynamically compiles an active character pool. Utilizing the `rand()` function, seeded by the current system time via `srand(time(0))` to prevent predictable sequence generation, the system constructs a randomized string of the specified length.

Strength Evaluation: The evaluation module assesses the password's entropy and complexity. It iteratively analyzes the string, flagging the presence of lowercase letters, uppercase letters, numeric digits, and special symbols. The tool then calculates an aggregate score and categorizes the password into one of three tiers: "Strong" (length ≥ 12 and 4 character types), "Medium" (length ≥ 8 and ≥ 3 character types), or "Weak" (failing the aforementioned conditions).

3. Conclusion

This automated utility provides an accessible and immediate solution for enhancing personal cybersecurity. By abstracting the complexity of true randomization and providing clear, actionable feedback on password strength, the tool encourages the adoption of secure authentication practices without requiring advanced technical knowledge.